

实验名称 实验5 基因识别的实现

一、实验内容及结果

1、修改所给出的 viterbi 算法代码，使其能够实现文件“基于 HMM 的基因识别实例.pdf”的例子

2、浏览下面的网站，查找适合 sequence.fasta 基因识别的方法，并进行基因识别蛋白

<https://esefinder.ahc.umn.edu/tools/ESE2/>

<https://services.healthtech.dtu.dk/services/HMMgene-1.1/>

<http://www.cbcb.umd.edu/software/glimmerm/index.shtml>

<http://exon.gatech.edu/GeneMark/>

TASK1:

代码:

```
import numpy as np
A = np.array([[ -0.097, -0.699],
               [-0.398, -0.222]])
B = np.array([[ -0.699, -0.523, -0.523, -0.699],
               [-0.398, -0.699, -0.699, -0.699]])

pai = np.array([[ -0.097],
                 [-0.699]])
O = np.array([[1],
               [2],
               [0],
               [0],
               [0],
               [0],
               [0],
               [0],
               [3],
               [1],
```

```

        [2]])
N = np.shape(A)[0]
T = np.shape(O)[0]
print('N=%d, T=%d'%(N,T))
delta = np.zeros((T,N))
psi = np.zeros((T,N))

for t in range(T):#[0,1,...,T-1]
    print('\n=====time t=%d'%t)
    if 0 == t:计算初值
        print('a0:', pai.reshape((1, N)), 'b0:', np.array(B[:,0[t]]).resh
ape((1, N)))
        delta[t] = np.add(pai.reshape((1, N)), np.array(B[:,0[t]]).reshap
e((1, N)))
        print('delta_t0:', delta[t])
        continue
    for i in range(N):
        print('\n*****i=%d'%i)
        print('delta[t%d-1]:'%t, delta[t-1], 'A[:,i%d]'%i, A[:,i])
        delta_t_i = np.add(delta[t-1], A[:,i])
        print('delta_t%d_i%d step1:'%(t,i), delta_t_i)
        delta_t_i = np.add(delta_t_i, B[i, 0[t]])
        print('delta_t%d_i%d result:'%(t,i), delta_t_i)
        delta[t,i] = max(delta_t_i)
        psi[t][i] = np.argmax(delta_t_i)
        print('delta:\n', delta)
        print('psi:\n', psi)

states = np.zeros((T,))
t_range = -1 * np.array(sorted(-1*np.arange(T)))
for t in t_range:
    if T-1 == t:
        print('time %d'%t, np.argmax(delta[t]))
        states[t] = np.argmax(delta[t])
    else:
        print('psi shape:', np.shape(psi))
        print('states shape:', np.shape(states))
        states[t] = psi[t+1, int(states[t+1])]
print('best state queue:', states)

path=[]
for i in states:
    if i == 0:
        path.append("N")

```

```

else:
    path.append("C")
print("状态为: ",path)

```

结果:

最优状态序列的概率数组:

```

delta:
[[-0.62  -1.398]
 [-1.24  -2.018]
 [-2.036 -2.337]
 [-2.832 -2.957]
 [-3.628 -3.577]
 [-4.424 -4.197]
 [-5.22  -4.817]
 [-5.914 -5.437]
 [-6.534 -6.358]
 [-7.154 -7.279]
 [-7.774 -8.2   ]]

```

最优状态序列的前导状态索引数组:

```

psi:
[[0. 0.]
 [0. 0.]
 [0. 0.]
 [0. 1.]
 [0. 1.]
 [0. 1.]
 [0. 1.]
 [1. 1.]
 [1. 1.]
 [0. 1.]
 [0. 1.]]

```

状态序列:

```
best state queue: [0. 0. 1. 1. 1. 1. 1. 1. 0. 0. 0.]
```

```
状态为: ['N', 'N', 'C', 'C', 'C', 'C', 'C', 'C', 'N', 'N', 'N']
```

TASK2:

1、不适用

sequence.fasta 中存储的是人类 DYSF 基因，属于 DNA 序列。网站 1 (<https://esefinder.ahc.umn.edu/tools/ESE2/>) 的注意事项中标注 ESEfinder 是一款 RNA 分析工具，故不适合识别 sequence.fasta。

Even though ESEfinder is an RNA analysis tool, only standard DNA notation is accepted (A, C, G, and T, not U). The program will ignore any character other than A, C, G, and T, including spaces, paragraph brakes and so on.

2、适用

网站 2 (<https://services.healthtech.dtu.dk/services/HMMgene-1.1/>) 是基于隐马尔可夫模型对基因结构的概率进行预测，能识别 DNA 序列中的基因的位置。

识别结果:

```
## gff-version 1
## date: Tue Jun 4 13:33:26 2024
## HMMgenel.1e (human model sim10gc.B.bsmod)

# SEQ: NG_008694.1_4939-238141 233203 (+) A:55221 C:55393 G:60396 T:62193
NG_008694.1_4939-238141 HMMgenel.1e firstex 439 526 0.981 + 1 bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e lastex 4082 4215 0.125 + 0 bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e CDS 439 4215 0.108 + . bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e firstex 13283 13373 0.914 + 1 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_1 27323 27378 0.941 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_2 28319 28410 0.997 + 2 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_3 49654 49759 0.996 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_4 58265 58361 0.717 + 1 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_5 60156 60361 0.908 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_6 62063 62225 0.427 + 1 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_7 67229 67344 0.819 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_8 72660 72786 0.846 + 1 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_9 74738 74841 0.986 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_10 81461 81529 0.992 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_11 81708 81751 0.935 + 2 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_12 85597 85679 0.873 + 1 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_13 97481 97596 1.000 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_14 98047 98160 0.970 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_15 99451 99628 0.979 + 1 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_16 100247 100371 0.750 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_17 102405 102511 0.998 + 2 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_18 108192 108384 0.982 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_19 110498 110653 0.832 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_20 114612 114778 0.969 + 2 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_21 116260 116374 0.954 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_22 116669 116774 1.000 + 1 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_23 117039 117181 0.784 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_24 120638 120811 0.918 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_25 122469 122618 0.508 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e exon_26 130829 130901 0.753 + 1 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e lastex 133068 133135 0.182 + 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e CDS 13283 133135 0.000 + . bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e firstex 136042 136126 0.314 + 1 bestparse:cds_3
NG_008694.1_4939-238141 HMMgenel.1e exon_1 136612 136728 0.732 + 1 bestparse:cds_3

# SEQ: NG_008694.1_4939-238141 233203 (-) A:62193 C:60396 G:55393 T:55221
NG_008694.1_4939-238141 HMMgenel.1e firstex 230373 230442 0.178 - 1 bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e exon_1 229685 229793 0.908 - 2 bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e exon_2 229000 229136 0.801 - 1 bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e exon_3 220596 220698 0.282 - 2 bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e exon_4 211927 212090 0.806 - 1 bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e exon_5 210308 210461 0.869 - 2 bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e exon_6 203702 203846 0.675 - 0 bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e exon_7 202812 203019 0.728 - 1 bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e lastex 182663 182949 0.338 - 0 bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e CDS 182663 230442 0.003 - . bestparse:cds_1
NG_008694.1_4939-238141 HMMgenel.1e firstex 148898 149105 0.069 - 1 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e lastex 148539 148624 0.058 - 0 bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e CDS 148539 149105 0.009 - . bestparse:cds_2
NG_008694.1_4939-238141 HMMgenel.1e firstex 113404 113469 0.046 - 0 bestparse:cds_3
NG_008694.1_4939-238141 HMMgenel.1e exon_1 108264 108372 0.119 - 1 bestparse:cds_3
NG_008694.1_4939-238141 HMMgenel.1e exon_2 106479 106600 0.822 - 0 bestparse:cds_3
NG_008694.1_4939-238141 HMMgenel.1e exon_3 98730 98750 0.432 - 0 bestparse:cds_3
NG_008694.1_4939-238141 HMMgenel.1e exon_4 94673 94780 0.781 - 0 bestparse:cds_3
NG_008694.1_4939-238141 HMMgenel.1e lastex 93162 93512 0.195 - 0 bestparse:cds_3
NG_008694.1_4939-238141 HMMgenel.1e CDS 93162 113469 0.000 - . bestparse:cds_3
NG_008694.1_4939-238141 HMMgenel.1e firstex 84460 84538 0.269 - 1 bestparse:cds_4
NG_008694.1_4939-238141 HMMgenel.1e lastex 72619 72659 0.189 - 0 bestparse:cds_4
NG_008694.1_4939-238141 HMMgenel.1e CDS 72619 84538 0.049 - . bestparse:cds_4
NG_008694.1_4939-238141 HMMgenel.1e firstex 35272 35327 0.582 - 2 bestparse:cds_5
NG_008694.1_4939-238141 HMMgenel.1e exon_1 24117 24138 0.148 - 0 bestparse:cds_5
NG_008694.1_4939-238141 HMMgenel.1e exon_2 12706 12989 0.395 - 2 bestparse:cds_5
NG_008694.1_4939-238141 HMMgenel.1e lastex 4267 4366 0.698 - 0 bestparse:cds_5
NG_008694.1_4939-238141 HMMgenel.1e CDS 4267 35327 0.036 - . bestparse:cds_5
```

```
## HMMgene1.1e (human model sim10gc.B.bsmod)

# SEQ: NG_008694.1_4939-238141 233203 (+) A:55221 C:55393 G:60396 T:62193
NG_008694.1_4939-238141 HMMgene1.1e firstex 439 526 0.981 + 1 bestparse:cds_1
NG_008694.1_4939-238141 HMMgene1.1e lastex 4082 4215 0.125 + 0 bestparse:cds_1
NG_008694.1_4939-238141 HMMgene1.1e CDS 439 4215 0.108 + . bestparse:cds_1
NG_008694.1_4939-238141 HMMgene1.1e firstex 13283 13373 0.914 + 1 bestparse:cds_2
NG_008694.1_4939-238141 HMMgene1.1e exon_1 27323 27378 0.941 + 0 bestparse:cds_2
```

以第一条结果为例，结果表明该基因第一个外显子有 0.981 的概率位于编码区域 `cds_1` 上。根据上图，选择概率高的识别结果，可以获知各编码区域的基因，进而翻译获取蛋白质序列。

3、不适用

网站 3 (<http://www.cbcb.umd.edu/software/glimmerm/index.shtml>) 是专门为真核生物开发的基因查找器，更适合拟南芥、水稻、恶性疟原虫等植物和微生物及其近似生物，所以也不适合对人类基因做出预测，对于 `sequence.fasta` 不适用。

an interpolated Markov model (IMM). The system has been trained for *Arabidopsis thaliana*, *Oryza sativa* (rice), and *Plasmodium falciparum* (the malaria parasite), and should work well on closely related organisms. See below for instructions on downloading the complete system including source code.

4、不适用

GeneMark

A family of gene prediction programs developed at
Georgia Institute of Technology, Atlanta, Georgia, USA.

网站 4 (<http://exon.gatech.edu/GeneMark/>) 是用于基因预测的网站，不适用于本例的基因识别。